

In this journal entry, I will share my thoughts and experiences on Module 6 and how to train a linear regression model and a logistic regression model. This lab was designed to teach me about each of the models and how to properly identify whether to use one model over the other. This module taught me about linear regression vs logistic regression, where linear takes input to give numerical outcomes, while logistic regression takes input to classify the input into a certain category on a familiar dataset, which was used in Modules 4 and 5 to practice one hot encoding, and now taking that data and making models with it.

This module taught me about 2 different types of ways to train a machine learning model. The first way is linear regression. This method is great for predicting how a number will change based on different inputs. When I first read it in the PowerPoint presentation, I thought about playing a video game where you can change your character's weapon or what buffs you receive. Based on multiple factors like weapon level and what items you may equip, your damage will increase; damage taken from a boss may reduce your damage. This is how linear regression works: your damage will be affected by each buff and debuff it receives. This model is trained by mean squared error. This is a great loss function to train linear regression, as if you have the same negative and positive numbers, they would just cancel out, but with this method they don't as they square the numbers, so any negative becomes positive.

The other method I learned in this lab is logistic regression. This is very similar to linear regression, where you're taking multiple inputs to get an output; however, linear will give a numerical output while logistic will put the output into different categories. Think about the video game analogy from before: take it as a team based game now instead of a single-player game. If you switch your character, your entire role has a chance to change, such as if you switch to a healer, your entire purpose in the game changes. Instead of being in the front line and dealing damage, you are now behind the main damage dealer, helping them survive. This is how logistic regression works: it takes character status and, based on this information, can predict what category they go in, if a character has a high attack stat and low to medium health points or has dealt most of the team damage, a logistic regression model can see this and predict with 90% probability that this character is a Main damage dealer.

In this lab, I used both models for the Titanic dataset. I used linear regression to predict fare and the inputs I took were age and Pclass. It returned 3364 for the squared error which

means when we take the square root of this number, my model is off by about 58 dollars. A question I had at this point was that, in the print statement, after it says that the model should only be off by 48 dollars, I had tried different inputs to get to this number, but it kept returning an error. I would like to know if this is my fault or if this is just a typo, and if it is my fault, I would like to learn what to do next time about a problem like this. For logistic regression, I used age, Pclass, and sex to determine if a passenger survived or not. I got a 74% accuracy rating. After I learned how both models work, I realized that the same dataset can be used for very different outputs depending on how your model works and what it's trying to achieve.\

These lab methods, like before, are based on careful consideration of what to use to train these models and what not to train these models with. If not picked carefully, this could waste resources and time. This could happen if you use the wrong inputs for your model and it won't have a high accuracy rate. I hope to learn about the other methods of machine learning later on in this course, and how they differ from these 2 models.